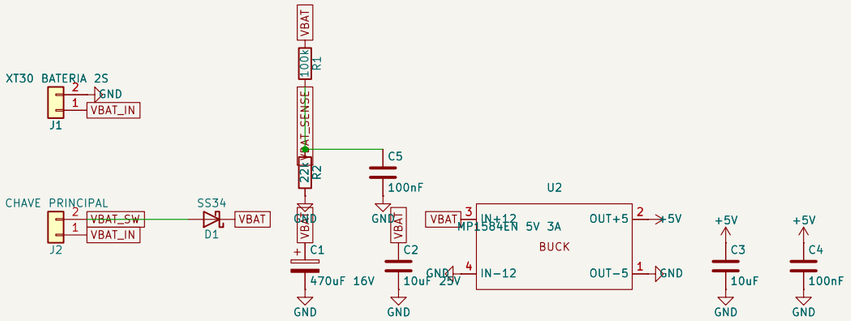
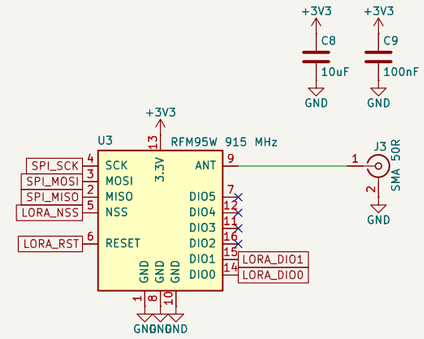
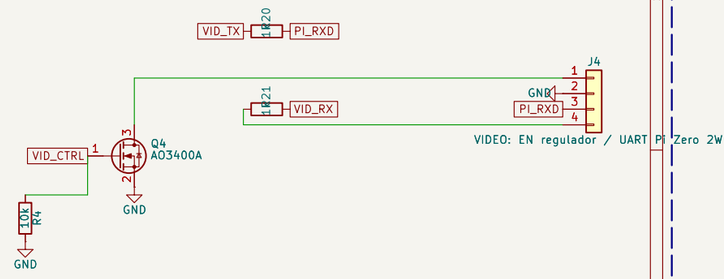
1. ALIMENTAÇÃO E MEDIÇÃO DA BATERIA



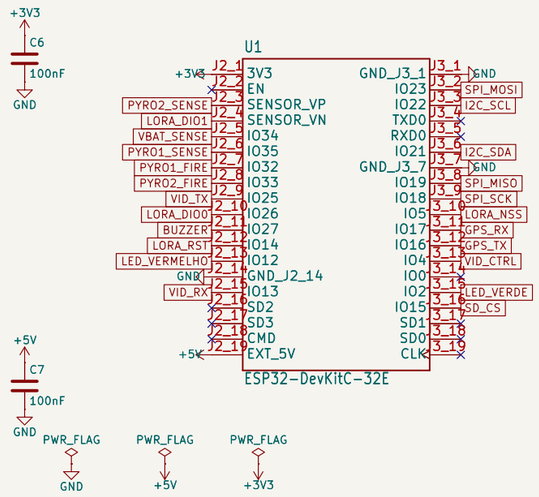
2. TELEMETRIA LoRa 917 MHz



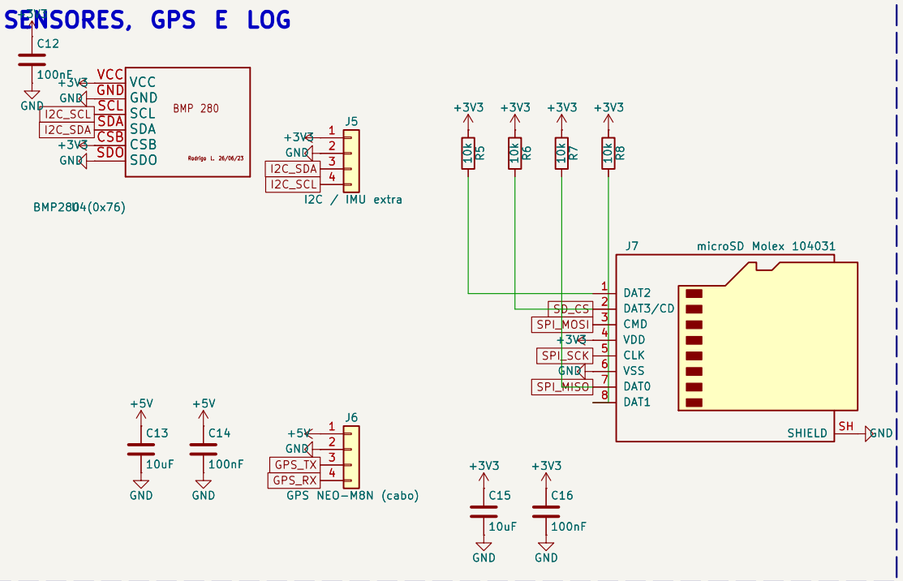
3. VÍDEO 2,4 GHz (Pi Zero 2W + SDR)



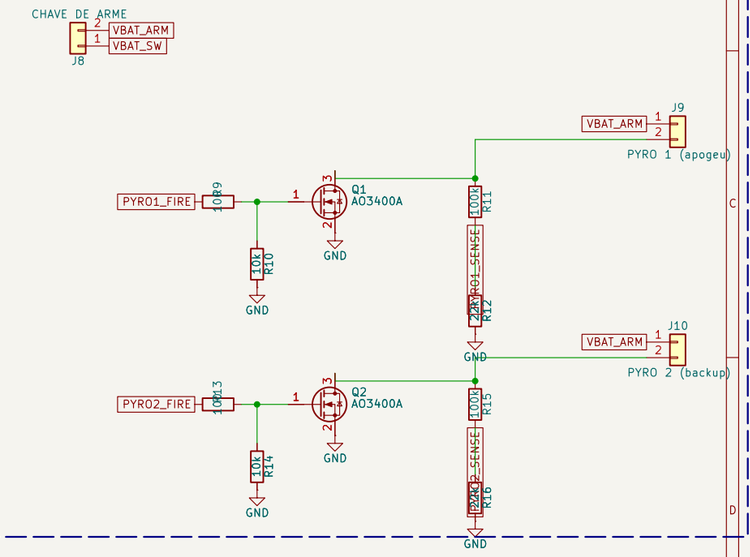
4. MICROCONTROLADOR



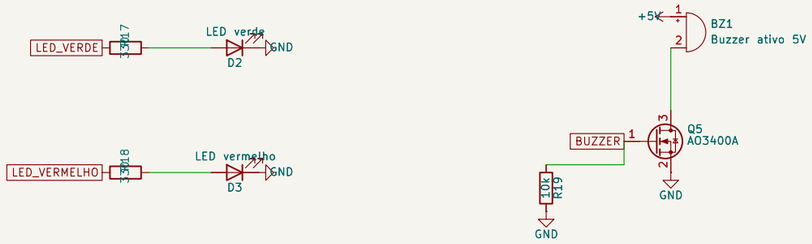
5. SENSORES, GPS E LOG



6. RECUPERAÇÃO (2 CANAIS PIROTÉCNICOS)



7. SINALIZAÇÃO



AVIÔNICA F-500 v2 rev.C-V (com vídeo) — foguete com apogeu ~500 m — UTFPR Rocket — segue o Guia de PCB (Payload 2026)

Corrente de pico (pior caso): ESP32 250 mA + RFM95W 120 mA + microSD 100 mA + GPS 45 mA + buzzer 30 mA + BMP/IMU 6 mA

=> +5 V: ~550 mA (2,75 W); entrada do buck a 80 %: 3,4 W / 6,4 V (2S descarregada) = 0,54 A; pulso pyro ~3 A

VÍDEO: bateria + regulador 5 V/5 A próprios (Pi 5 V/2,5 A + BOB/SDR + FEM ~0,5 A). Aqui só passam VID_OFF e UART (mA)

H1 M3 H2 M3 H3 M3 H4 M3

Esquemático em blocos conforme o Guia de PCB (correções Payload 2026)

UTFPR Rocket

Sheet: /

File: avionica_f500_v2_video.kicad_sch

Title: Aviônica F-500 v2 rev.C-V — telemetria p/ tracker + vídeo 2,4

Size: A3	Date: 2026-11-09	**Rev: C-V**
KiCad E.D.A. 10.0.1		Id: 1/1